FarSightBERT: Enhanced Embeddings for Long-Term Disease Prediction

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***Abstract*—???**

***Index Terms*—Scientific writing, Typesetting, Document creation, Syntax, Clinical decision support systems, Disease prediction, Healthcare analytics, ICD-9 code group prediction, Precision medicine**

I. Introduction

The FarSight paper [1] addresses the problem of predicting diseases using unstructured clinical notes. The premise of that paper was that while most models use electronic health records (EHR) data to model on, valuable information exists in unstructured clinical notes. The authors Gangavarapu et al develop a long-term aggregation mechanism to capture onset of diseases from earliest recorded systems.

The core contributions of the FarSight were:

- Novel aggregation mechanism to identify earliest signs of disease.
- The fact that unstructured nursing notes can be a valuable data source.

We try to extend these contributions by ???

II. Scope of Reproducibility and Novelty

In this study, we reuse the original data aggregation mechanism developed in FarSight. We extend the original paper with 2 novelties that we shall explain below in detail.

A. Bidirection Contextual embeddings (BERT)

The authors in the paper use various embedding techniques, including Doc2Vec and multiple variants of Non-negative Matrix Factorization (NMF) based on Bag-of-Words (BoW) and Term Weighting (TW), with and without Semantic Coherence (SC).

We differ in our approach by using a pre-trained BERT model, Bio_ClinicalBert [2]. Bio_ClinicalBert is based on BioBERT [3] which itself is a domain-specific version of BERT [4] pretrained on biomedical literature such as PubMed abstracts and PMC full texts. Bio_ClinicalBert extends BioBERT by pretraining on the large unstructured notes in the MIMIC-III [5] dataset and is publicly available on HuggingFace [2].

BERT being an encoder-only model (which gives it bidirectional context) allows the model to build better representations of vocabulary and structure of clinical narratives making it well suitable for downstream tasks such as the one in this study.

We hypothesize that having better embeddings using BERT will help counter many problems faced by the authors when they were experimenting with different modeling approaches. To list a few: BERT will be able to better capture semantic relationships compared to BoW and Doc2Vec, handle complex medical language and be able to leverage a strong foundation from its pre-trained knowledge.

B. Mixture of Expert (MoE) models

We further extend this approach by using some of the models introduced in FarSight with a powerful extension call MoE [6]. The FarSight paper identifies several challenges in dealing with clinical notes, including their longitudinal aspect, heterogeneity, voluminous content, and complex structure. The idea of using MoE is inspired to handle heterogeneity by specialization.

Each model (expert) optimizes on it's own inductive biases and hence, we expect each model to specialize on different aspects of the data. For e.g., ConvLSTMs may be better at capturing temporal pattern in notes while ConvNet may be better at identifying specific keywords in shorter, acute-event notes.

The MoE learns to combine strengths of different neural architectures to make the best overall prediction. Furthermore, this modeling approach gives potential for interpretability as we can analyze which architecture the gating network relies on most for different type of notes. This can provide insight into which architectural biases can be most useful for different clinical scenarios.

III. Methodology

A. Dataset description

MIMIC-III v1.4 [7] is a publicly available, de-identified database comprising detailed health-related data from over 40,000 patients admitted to critical care units at the Beth Israel Deaconess Medical Center in Boston, Massachusetts, between 2001 and 2012. The MIMIC-III v1.4 database consists of 2,083,180 note events out of which 223,556 are nursing notes from 7,704 distinct patients. We model on these nursing notes which is a huge a corpus of 5,244,541 sentences, 79,988,065 total words and 715,821 unique words.

B. Dataset Cleaning and Preprocessing

a) *Cohort selection:* Our cohort selection approach is same as the one illustrated in the FarSight paper. We do the following: (1) we filter out neonates (age < 15), (2) keep only first ICU admissions for each MIMIC-III subject and discard later admissions, (3) identify and filter out any nursing notes with clerical error attribute, and (4) remove duplicate patient records.

The authors do (1) to maintain consistency in benchmarking with respect to related works, (2) was done because the authors found that for 94% of patients, diagnostic code groups in first admissions overlapped with those in later ones. Furthermore, using only first admissions helps avoid statistical complications that arise from having multiple admissions from same patient, this also helps in decreasing computational requirements without significant loss of information.

b) *Text preprocessing:* Our text preprocessing steps differs a little bit compared to the FarSight paper. Once the cohort selection has completed, we preprocess the text data in the nursing notes. We do the following: (1) the text is split into individual tokens using NLTK tokens, (2) common stopwords are eliminated using NLTK English stopword corpus, (3) we remove any punctuation marks, (4) text is converted to lowercase, (5) stemming and lemmatization is applied, and (6) tokens appearing in fewer than 10 nursing notes are eliminated.

We differ in our approach from the original paper by not doing Medical abbreviation disambiguation. It was not possible for us to find the CARD-2 framework used in the in original implementation. Furthermore, while this could not be evaluated, we leave this load on the Bio_ClinicalBert to give contextualized embeddings that take care of this for us implicitly.

c) *BERT embeddings (Clinical Feature Modeling):*

d) *FarSight Data Aggregation:* We apply FarSight's aggregation mechanism to map each nursing note to all ICD-9 diagnostic code groups observed in that patient's admission. This enables detection of disease onset with early symptoms before any formal diagnosis.

e) *ICD-9 code grouping:* Mapped ICD-9 diagnostic codes into 19 distinct diagonistic groups based on code ranges defined in [8]. ICD-9 code range of 760-779 corresponding to neonates (age < 15) are not part of our cohort and excluded in this study. Furthermore all reference and supplemental V-codes are grouped into same code group to lower computational complexity to train.

C. Model description

We use many different architectures in the FarSight paper.

IV. Training

A. Computational implementation

B. Training details

V. Evaluation

VI. Results

VII. Discussion

VIII. Authors Contributions

1) Nikhil Kapila:
   - Created data pipeline.
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2) Tejas Rathi
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